

Quiz 4
CSE471: System Analysis and Design
SET-A

Name:

ID:

Total Marks: _____ / 20

Time: 25 minutes

Data Store Design & Program Design (CO2)

Q1. You should show the proper units for all the questions, detailed calculations, and/or formulas.

Attributes	Average Size (bytes)
Name	2
Id	2
Address	4
SSN	4
Email	7
Designation	8
Mobile	9
Salary	13

- a. Find the **Record Size** in **bits**. [2]

$$\begin{aligned}\text{Record Size} &= \text{Summation of all the attributes' average size} \\ &= ((2 + 2 + 4 + 4 + 7 + 8 + 9 + 13) * 8) \text{ bits} \\ &= 392 \text{ bits}\end{aligned}$$

- b. The system's overhead will be 0.3. Calculate the **Total Record Size** in **bits**. [2]

$$\begin{aligned}\text{Total Record Size} &= \text{Record Size} + (\text{Record Size} * \text{overhead}) \\ &= 392 + (392 * 0.3) \text{ bits} \\ &= 509.6 \text{ bits}\end{aligned}$$

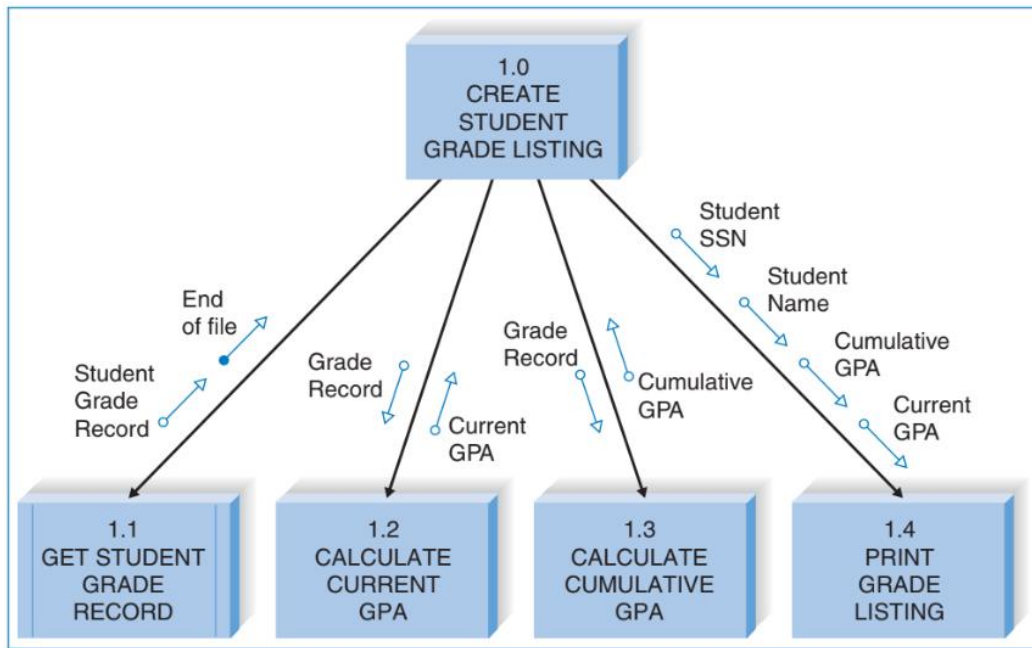
- c. The initial table size is 50,000. Calculate the **Initial Table Volume** in **bits**. [2]

$$\begin{aligned}\text{Initial Table Volume} &= \text{Initial Table Size} * \text{Total Record Size} \\ &= 50000 * 509.6 \text{ bits} \\ &= 25,480,000 \text{ bits}\end{aligned}$$

- d. Calculate the **Final Table Volume** in **bits** after 36 months; quarterly growth is 3000. [4]

$$\begin{aligned}\text{Final Table Volume} &= \text{Initial Table Volume} + (\text{Growth per unit} * \text{Total Record Size} * \text{total time}) \\ &= (25,480,000 + (1000 * 509.6 * 36)) \text{ bits} \\ &= 43,825,600 \text{ bits}\end{aligned}$$

Q2.



- a. How many control and subordinate modules are there? [2]

Ans: Control – 1, Subordinate - 4

- b. What kind of module is 1.1? [2]

Ans: Library

- c. Which unique control and data couples do you see? [2]

* Unique means appearing only once [Student Grade Record = Grade Record]

Ans: Control – End of file, Data – Student SSN, Student name.

- d. What are the fan-in and fan-out numbers for 1.2 and 1.3? [2]

Ans: 1.2 – Fan-in: 1, Fan-out: 0, 1.3 - Fan-in: 1, Fan-out: 0

- e. How many methods should the module 1.0 call at max? [2]

Ans: 7